Plexus Discovery

A specification for an autonomous mechanism-search campaign on Okuda tubulation

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Abstract

This is a build specification, not an essay. It defines an agentic loop that searches for the *mechanism* behind a morphogenetic phenotype — as a composition of Plexus operators — and that is designed to run unattended on a GPU partition for weeks without descending into the failure mode that consumed our last forty-four rounds.

The target system is Okuda tubulation: a closed epithelial shell that grows a tube. We already make a tube; we make it *by the wrong mechanism*, and we know this because relaxing our tube toward mechanical equilibrium destroys it (§1). That gives the campaign a sharp, falsifiable objective rather than a resemblance target.

Section 2 states what machinery already exists and transfers. Section 3 defines the Okuda composition space concretely — operators, ports, stages, legal edits. Section 4 diagnoses why searches like ours go into rabbit holes, from our own record with numbers. Section 5 is the core of the document: an explicit control law with a batch–triage–truncate–starve–freeze–terminate cycle, so that budget is *taken away* from unproductive branches automatically. Section 7 assigns the work to agents, borrowing orchestration from two multi-agent systems published together in *Nature* in July 2026 while deliberately diverging from their adjudication. Sections 9–10 define the artefacts, including the causal descriptions and the trigger that hands a problem to the differentiable fit. Section 11 is a pre-flight checklist of eleven defects that must be closed first — five of them found while writing this document.

Contents

1 The question the campaign answers

1.1 Not "can we make a tube"

We can. A tip-tracking activation driver on a growing, dividing vesicle produces a long, thin, constant-diameter tube with a length-to-diameter ratio above four, and its diameter tracks the morphogen diffusion coefficient — a qualitative reproduction of the published law. That question is closed.

1.2 The question is which mechanism holds it open

Round 41 of that campaign asked what happens if the tissue is pushed toward the published *regime* — Okuda's model is explicitly quasi-static, relaxing to force balance between growth events. The answer was decisive:

> **Relaxation destroys our tube.** More relaxation collapses it back toward a sphere. Our tube is held open by an explicit outward force; remove the force and the shape energy pulls it in. The published tube is an *equilibrium* shape produced by localised proliferation.

The mechanical signature agrees: force and pressure are concentrated entirely in the protrusion, the body is idle, and division and migration pile up at the same tip. That is what a forced protrusion looks like. A grown one shares load with the body.

So two mechanisms produce the same picture, and the picture is exactly what cannot tell them apart. This is the general statement of why parameter search cannot find a mechanism, and it is stronger than the usual one: *the search optimises resemblance, and resemblance is the quantity blind to the distinction that matters.*

1.3 The campaign objective, stated falsifiably

> *Find a composition of Plexus operators that produces a sustained tube as a* ***growth-driven quasi-static equilibrium*** *— one that survives relaxation, shares load between tube and body, and requires no explicit extrusion force — or establish that no composition in the searched space can, and say precisely which capability is missing.*

Two subordinate questions inherit from the record and must be re-tested rather than assumed:

1. **Can patterning be emergent?** A late round concluded that the activator–inhibitor system floods the growing domain and cannot maintain a tip-riding peak, so the driven tip signal is necessary. That is an impossibility claim, and it was made *before* the precondition checks of §11 existed. It is exactly the class of claim a silently inert operator manufactures. It re-enters the campaign as **Open**.

2. **Does per-cell three-dimensional volume change the answer?** The mid-surface model gives a cell a wedge volume measured from the shell centre; the monolayer implementation gives it a true volume $A \cdot h$ with apical and basal surfaces, and with it a bending rigidity that is emergent rather than imposed. The first attempt to wire it in produced spikes, and the diagnosis on record is that division still triggers on the wedge volume, so activated cells inflate instead of dividing. This is a concrete, testable entry point, not a mystery.

1.4 The metric that makes the objective measurable

The objective above is useless without an observable that separates a forced tube from a grown one. No metric in the existing bank does. We therefore define one, and it is cheap:

The quasi-static test. Take a run's final state; continue the simulation with *growth and any driver switched off* and only mechanics and reconnection active; measure what fraction of the protrusion survives.

$$Q = \frac{\text{aspect after relaxation}}{\text{aspect before relaxation}}.$$

A forced tube collapses ($Q \to 0$). An equilibrium tube persists ($Q \to 1$). Alongside it we record the pressure ratio between tube and body, which should approach unity for a grown tube and is presently around three.

Q is the campaign's primary discriminator. It is defined before the search starts, it is falsifiable, and it is the direct operationalisation of the round-41 finding. Authoring it is an instance of the discipline this whole document is built around: *when the eye sees a distinction no number captures, the missing number is the deliverable.*

2 What already exists and transfers

A working three-loop discovery system exists — roughly 1,100 lines — and has run unattended for six rounds on embryonic gland branching, producing 1,216 archived runs and one conclusion that a later, better measurement overturned. It is the substrate for this campaign. This section records what transfers unchanged, what must be replaced, and what must be added.

2.1 Transfers unchanged — the three structural guarantees

1. A composition is a typed graph, and its identity excludes parameters. A candidate mechanism is operator nodes plus typed connections between output ports and input slots. A morphogen may be routed to a cleft `source`, a growth `gate`, a chemotaxis `field` or an anisotropy `axis` — four *different mechanisms*, not four settings. Identity is a hash of the structure only:

```
def comp_hash(composition):
    # operators + typed connections ONLY -- no parameter values, no seed
```

Therefore **a change of numbers provably cannot register as a new hypothesis**. This single line is the structural fix for what happened over rounds 1–30, and it is why the campaign can be trusted to count its own progress honestly.

2. Evidence and interpretation are separate objects. The rule is enforced in code, not in prose:

$$\text{simulation} \longrightarrow \text{RunRecord} \longrightarrow \text{knowledge} \qquad \text{(never simulation} \longrightarrow \text{knowledge).}$$

A `RunRecord` carries the raw facts of one simulation — composition, parameters, seed, backend, initial condition, parent, and the single edit that produced it — and is *immutable*: assigning to a raw fact raises. Measurements attach as *versioned, append-only* analyses, so `metric_v2` sits beside `metric_v1` and neither is lost. The archive is append-only, and run identity is a content hash of exactly the reproducible inputs, so a crashed campaign resumes without duplicating or losing evidence.

This is what let a new observable, in the gland campaign, **re-score archived runs and change a mechanistic conclusion without re-simulating anything**. It is the most valuable property the system has.

3. One legal edit at a time, stage-gated. Four moves exist — add, remove, connect, disconnect — each returning a new graph together with the edit that produced it. Every run records its parent and its mutation, so the archive is a causal tree, not a bag of configurations. Stage gating keeps the legal move set small and interpretable rather than an enumeration of 2^N subsets.

2.2 Must be replaced

Component	Why it must change
Operator vocabulary	The existing one is phase-field gland vocabulary. Okuda needs its own (§3).
Frontier expansion	Breadth-first with a node cap, spending budget evenly on good and hopeless branches. Replaced by the control law of §5.
Trajectory storage	Only the final frame is persisted. Fatal here: every decisive observable in the tube record is temporal.
Verdict thresholds	Hard-coded constants on a four-sample estimate. Replaced by explicit, recorded, interval-based promotion rules.
Impossibility reasoning	Currently an `if/elif` chain that guesses a reason from absent operators. Replaced by an earned claim (§11, defect 8).

2.3 Must be added

The supervisor, the cheap-tier triage, the proximity starvation rule, the watcher, the causal interpreter, and the artefact pipeline — all specified below.

3 The Okuda composition space

The space is defined by an operator vocabulary with typed ports, a stage gate, and a seed. All operators below are already registered and exercised; none is hypothetical.

3.1 Vocabulary

Stage	Operator	Kind	Ports
1 — substrate	`seed_mesh_3d` / `load_mesh_3d`	structural	—
	`shape_energy_3d` *(default | monolayer)*	lateral	`a0` slot
	`reconnect_t1_3d`	rewire	—
2 — growth & topology	`vesicle_growth`	structural	—
	`morphogen_growth_3d` *(conserve_amount)*	structural	`gate` ← morphogen
	`divide_3d` *(hertwig | orient_iface)*	structural	`axis` ← morphogen
	`extrude` *(the forcing term)*	lateral	`site` ← morphogen
3 — patterning & feedback	`cell_geometry_3d`	aggregate	→ geometry
	`cell_adjacency`	rewire	→ adjacency
	`cell_diffuse`	lateral	needs adjacency
	`cell_react` *(gray_scott | gierer_meinhardt | brusselator)*	lateral	→ morphogen
	`cell_rd_seed` *(cone | tip | spot | none)*	structural	→ morphogen

Two design choices in that table are load-bearing.

Extrusion is an operator node, not a parameter. The whole round-41 finding is "ablate the forcing and the tube collapses". If the forcing lives inside a configuration file it can only be turned down; if it is a node in the graph, the necessity test **ablates it automatically**, on every composition, as part of the standard protocol. The campaign's central question thereby becomes a routine measurement rather than a special experiment.

Three reaction kinetics are three implementations of one contract. `cell_react` already resolves to `gray_scott`, `gierer_meinhardt` and `brusselator`. Swapping them is an implementation choice under a single biological contract, which is what the language claims and here is simply true. The search can therefore ask "which kinetics" without changing the composition's identity, and ask "with or without reaction at all" as a genuine structural edit.

3.2 Seed and legal edits

The seed is Stage 1 only: a mesh, a shape energy, and reconnection — a relaxing vesicle that does nothing. Every subsequent mechanism is reached by one legal edit. Crucially the search visits the degenerate corners on the way (growth with no patterning, patterning with no growth, mechanics alone), and those are where impossibility results come from.

3.3 Phenotype and the metric bank

Quantity	**Reads**	**Answers**
`aspect`, `tube_len`, `n_tubes`	shape	is there a tube?
`Q` **(quasi-static)**	shape after driver-off relaxation	**is it grown or forced?**
`p_tube/p_body`	differentiable stress field	is load shared?
`hollow` (tube-aware), `area_CV`	mesh integrity, uniformity	is the wall real?
`red_frac`, `tip_act`	activator confinement, activator–radius correlation	does patterning ride the tip?
`n_cells`, `shape_index`	proliferation, cell rounding	is it well-formed?

Each is computed *per frame*, not only at the end — see defect 7.

4 Why searches like ours go into rabbit holes

Every defect in §11 is fixable in days. The harder problem is that a search can be free of bugs and still waste months. Our own record supplies the diagnosis, and the pathology has five separable components — worth separating because each needs a different countermeasure.

1. **Depth-first drift.** Three separate rounds tested, in substance, "the same thing but more": run longer, activate a bigger patch, make the wall thicker. All three were falsified. Each was a locally reasonable next step from the configuration in front of us. None was a new hypothesis. A search that always expands its current best node does this forever.

2. **No terminal-state rule.** Nothing ever said "this branch is finished". Rounds continued because there was always another knob. Stopping was a human losing patience, which is not a criterion.

3. **Near-duplicate proliferation.** Thirty rounds explored perhaps four genuinely distinct ideas. Without a distance between hypotheses, twenty variants of one idea consume twenty times the budget of one idea — and their collective failure is then misread as strong evidence against the idea, rather than as one observation repeated.

4. **The eye and the number disagreeing, undetected.** Two of the most confident intermediate conclusions were measurement artefacts: a "97% of cells inverted" result produced by pairing each frame's coordinates with a different frame's connectivity, and a "101 chemical spots" count where the eye saw three, because the threshold sat inside the noise. Both were

caught by looking at a picture. Neither by a number. In a supervised session a human notices; in an unattended campaign nobody does.

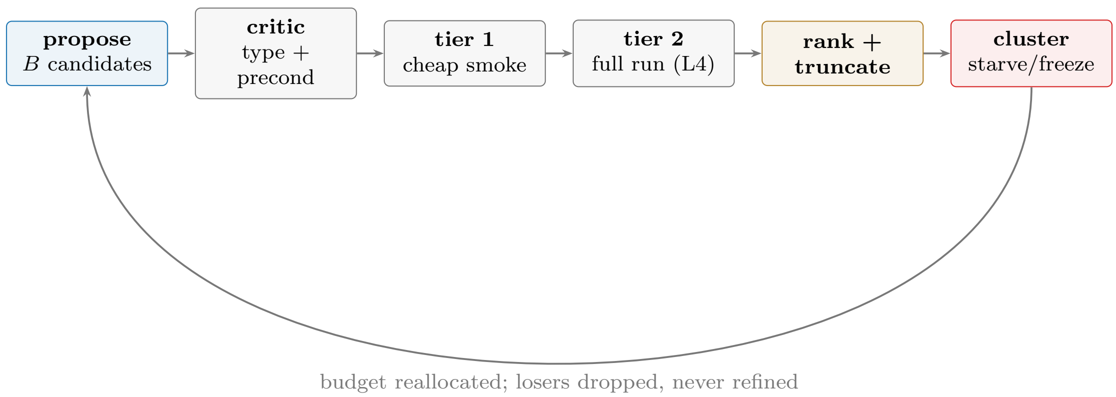
5. **The human as sole goal-holder.** An agent elaborates whatever is in front of it and cannot detect its own drift. Every course correction on record — "we are meant to be testing the growth-driven mechanism, not tuning this extrusion" — came from outside the loop.

The rest of this document is, in effect, five countermeasures.

5 The control law

This is the core of the specification. The governing idea is that budget must be *removed* from unproductive branches automatically, because no one will be watching.

5.1 One round



Stage	Keeps	Rule
Propose	$B = 24$	One legal edit each, drawn across *distinct* clusters, not all from the current best.
Critic	—	Free rejection: ill-typed edit, unmet precondition, structural hash already evaluated. No simulation.
Tier 1	8	Small mesh, short horizon, seconds each. A gate, never evidence.
Tier 2	8	Full run on the partition, full metric bank, per-frame.
Rank	—	By measured metrics. Pairwise with a Bradley–Terry–Luce aggregation where a scalar order is ill-defined; never win/loss tallies, which carry position bias.
Truncate	3	**Losers are dropped, not refined.** They stay in the archive as evaluated evidence and are never re-expanded.
Starve	—	Within-cluster competition; a cluster's members compete against each other, so a family of near-identical ideas exhausts one budget, not twenty.
Freeze	—	A cluster with $> K = 6$ evaluations and no improvement in its best score is frozen; its budget is reallocated the same round.

5.2 Termination and escalation

The supervisor owns the campaign configuration — objective, success criteria, stopping rule, budget — and workers cannot amend it. Each round it computes:

```
if objective_met(ledger):            terminate("objective")
if all_clusters_frozen():            escalate()      # widen stage gate, or request an operator
if dry_rounds >= 2:                  escalate()      # no new cluster AND no best-score gain
if budget_spent >= budget:           terminate("budget")
```

Escalate is the important branch, and it is what a human did by hand last time. It means: the current space is exhausted, so either open the next stage gate, or record an **operator request** — a mechanism the proposer wanted and could not express — and move on. Escalation is logged and is a deliverable (§9).

5.3 Rules enforced in code, not recommended in prose

- **A change of numbers is never a new hypothesis** — guaranteed by the structural hash.
- **No manual parameter tweaking outside the loop.** If a configuration needs hand-tuning to work, that is evidence of a missing operator, not of a missing afternoon.
- **No silent caps.** Any round that truncates, samples, or drops candidates says so in the log with counts. A silent truncation reads afterwards as complete coverage.
- **Structural limitations are recorded once and never retried.**
- **Per-round compute is bounded.** Depth is traded against basin width inside a fixed budget, and the trade is recorded. The current driver deepens every four rounds with no

ceiling, which silently converts a campaign of many cheap experiments into a few enormous ones.

6 The scientific protocol: hypothesis first

The control law of §5 says how budget is allocated. This section says what makes an allocation an *experiment* rather than a sample. One rule, enforced in code:

A candidate is not run until its prediction has been recorded.

Every candidate carries, before it reaches the cluster: a falsifiable *claim*, the single *metric* that decides it, an explicit *prediction* on that metric, a *rationale*, and the Grounder's *citations*. A hypothesis without a prediction is refused at construction.

6.1 Two intents, and why the mixture matters

Each batch is allocated between two kinds of edit:

Confirmatory	an edit the current causal map predicts *will* work. Consolidates the map. Individually low-information; collectively what makes the map usable.
Adversarial	an edit predicted to *break* the current best explanation. High-variance; this is where surprise lives.

Because the prediction precedes the run, every outcome falls into a 2×2:

	confirmed	**refuted**
confirmatory	consolidates (low information)	**surprise** (high information)
adversarial	**surprise** (high information)	breaks as expected (low information)

A batch of pure confirmation (100/0) only ever confirms what we already believe: near-zero information. A batch of pure falsification (0/100) only ever probes the improbable, which mostly breaks as expected and never consolidates a working map. The productive operating point is a deliberate minority of attacks on the current understanding — roughly 70/30, and at times 50/50 when the map is young.

6.2 The 70/30 balance is a setpoint, not a quota

This is the part that makes the discipline machine-enforceable. Define the *surprise rate* of a round as the fraction of resolved hypotheses whose outcome contradicted its prediction. Then the Supervisor closes a loop on it:

surprise	**diagnosis**	**correction**
< 0.10	drifted to 100/0: confirming what we already believe	push **adversarial**
≈ 0.30	the productive band	hold 70/30
> 0.50	drifted to 0/100: everything breaks, no map accumulates	push **confirmatory**

A quota cannot notice that its questions have become trivial; a setpoint on the prediction-error rate can. This is the mechanism by which the loop avoids both failure modes the hand-run campaign fell into — a long stretch of confirmatory tuning where nothing surprised us, followed by a stretch of wild attempts that mostly broke as expected.

6.3 The knowledge document

`knowledge.md` is append-only and written in the order the science happened: the hypothesis and its grounding, then its prediction, then what occurred. A reader can audit what was believed *before* the evidence arrived — which is the only way to tell a prediction from a post-hoc rationalisation. Verdicts move between the four classes of §9; a claim's history is extended, never rewritten. Surprises are flagged, because they are the entries a returning human should read first.

7 The agents

Two systems published together in *Nature* (July 2026) solve much of the orchestration problem for literature-driven hypothesis generation with wet-lab validation: Co-Scientist, a supervisor-coordinated coalition with an Elo tournament, a proximity graph and a meta-review that appends learned patterns to other agents' prompts; and Robin, which batches and truncates, ranks by an LLM-judged pairwise tournament aggregated with a Bradley–Terry–Luce model, and runs eight independent analysis trajectories reconciled by consensus.

Our setting differs in one decisive respect. **Their experiment is scarce and ours is cheap.** A wet-lab assay costs weeks, so ranking by argument is rational. Our simulation costs seconds and is measured by code. We therefore take their orchestration, diverge on their adjudication, and add one agent neither has.

Agent	Role	Provenance / divergence
Supervisor	Holds objective, criteria, stopping rule, budget. Computes round statistics; decides allocation, freezing, escalation, termination. Never does the work.	Co-Scientist's Supervisor, whose statistics likewise determine whether a terminal state is reached.
Grounder	Holds `plexus2.tex` (the language contract), `okuda.pdf` (the reference model) and `okuda_corpus.md` + `papers/` (the literature). Answers mechanism questions and attaches citations.	Robin's literature agents. Co-Scientist's ablation showed that giving the reviewer search suppressed plausible-but-false hypotheses.
Proposer	Generates a batch of one-edit candidates across distinct clusters.	Co-Scientist's generator. *Divergence:* our space is typed and executable, so illegal proposals are impossible by construction rather than rejected on review.
Critic	Free pre-compute triage: types, preconditions, duplicates.	Co-Scientist's reflection, reduced to what can be decided without simulating.
Referee	Ranks a batch.	Robin's tournament shape and BTL aggregation. **Divergence: the judge is the metric bank, not a debate.**
Proximity	Clusters compositions by structural distance for within-cluster competition.	Co-Scientist's proximity graph. Ours is *exact* for identity (the hash) and approximate only for similarity.
Analyst $\times N$	N independent readings of the same trajectory, reconciled by consensus; disagreement is itself recorded.	Robin's eight parallel analyses. Aimed squarely at our two measurement artefacts.
Watcher	Reads the rendered movie; describes the dynamics; flags divergence between picture and score. Blocks promotion on disagreement.	**Neither system has this.** Both are text- and table-bound. Every decisive call in our record came from watching a movie.
Interpreter	Writes the causal description (§9) as a versioned analysis.	Co-Scientist's overview synthesis, specialised to mechanism.
Meta-review	Distils recurring patterns and appends them to proposer and critic prompts.	Co-Scientist, which calls this feedback propagation without backpropagation — no fine-tuning.
Metrologist	Owns the validation ladder, the instrument gate and *substrate semantics* (units, clocks, per-call vs per-frame). Certifies that evidence may be admitted — and **retracts knowledge already recorded** when a foundation defect is found.	**Neither system has this.** Their measurement is an external wet-lab assay; ours is code we wrote.
Engineer	Proposes a patch to the substrate when the Metrologist quarantines it. Writes to a *different artefact* than the campaign's evidence, and cannot admit its own fix.	Deliberately outside the loop (see the boundary below).

The Grounder, specifically. It is worth being precise about this agent because it is the one whose absence was most expensive last time. The Grounder holds three bodies of text — `paper/plexus2.tex` (what the language permits), `papers/okuda.pdf` (what the reference model actually does), and `papers/okuda_corpus.md` together with the vendored `papers/` corpus (what the field has already established) — and it is called at exactly three points:

1. when the **Proposer** needs a mechanism grounded before proposing it;
2. when the **Supervisor** gates a hypothesis, before Discovery is allowed to propose;
3. when an **operator request** is filed — does the literature already name this mechanism, and under what conditions does it hold?

This is the agent that produced the Gierer–Meinhardt unlock by hand last time. The decisive question — *which reaction–diffusion regime does Okuda actually use?* — was answered by reading the paper, not by sweeping parameters, and it redirected the entire campaign. Making that a scheduled call rather than a lucky afternoon is the point.

Two further borrowings worth naming. Whatever proposes must not be what judges — Robin used different models for generation and adjudication, and the separation is free. And Robin found their orchestrator "almost always called tools in the same order", so they replaced the agentic controller with a deterministic notebook. We follow that: **the control law of §5 is a deterministic script**, and language models are called only where judgement is genuinely required — proposing, watching, explaining.

One thing neither paper needs. Their measurement is an external assay. Ours is code written by the same system being judged, so "the metric lied" is a failure mode unique to us, and we have suffered it twice. Hence the instrument gate of §11.

8 Knowledge certification, and who may touch the substrate

If the foundations are wrong, so is the knowledge.

Every other agent in §7 produces or judges *evidence*. None of them is responsible for the thing evidence is measured *with*. That gap is not hypothetical: it is how the defect below was found — by a person reading code, not by any designed role.

8.1 A worked example of why the role must exist

`divide_3d` counts `min_cycle` and `max_cycle` in *division-calls*, and `max_div` / `max_div_frac` are *per-call* throttles. The archived configurations passed `every: 2`, which the engine gated *and* the operator's own private counter gated — so the operator body executed once every **four** frames. Correcting the clock therefore multiplied the wall-clock meaning of every per-call quantity by four, and the same recipe that had produced a tube of aspect 7.5 produced one of aspect 3.2.

Two further consequences, each of which a casual fix would have missed:

- The hand-tuned values had been adopted as *vocabulary defaults*, on the reasoning that "a default composition should start where our evidence is". After the clock fix that reasoning inverted: those defaults now encode a θ tuned for the wrong clock, so *every* composition the loop generates would have started biased — silently.
- Rescaling the obvious knob was not enough. Since cap $= \max(\texttt{max\_div}, \texttt{max\_div\_frac} \cdot n_F)$, the absolute floor *dominates* at realistic cell counts, so rescaling only `max_div_frac` was **entirely masked**. Both had to move.

Because the factor is exact, the correction is analytic rather than a re-tuning sweep: rescale the per-call quantities and the fix becomes *behaviour-preserving by construction*, verified identical to the archived budget at every cell count. Any later change in phenotype is then attributable to the change that caused it, rather than to the clock.

8.2 The Metrologist

The Metrologist certifies that evidence may be admitted at all. It owns the validation ladder, the instrument gate, and — the part no other role covers — *substrate semantics*: units, clocks, and the distinction between per-call and per-frame quantities. Its distinguishing power is that it may act *backwards*:

> When a foundation defect is found, the Metrologist issues a **retraction**: every claim whose evidence depends on the defective semantics is moved back to *Open*, with the defect and its scope recorded.

In an append-only ledger a retraction is a new record, never an edit. The history must show what was believed, on what evidence, and why that evidence was later withdrawn — otherwise a reader cannot distinguish a claim that survived scrutiny from one that was quietly repaired.

8.3 The boundary: who may modify the substrate

> **No campaign agent may modify the substrate.** An agent that can rewrite the simulator can make any hypothesis true.

The separation is therefore structural, not procedural:

Metrologist	*detects and quarantines.* Halts admission of evidence, marks affected claims Open-pending-recheck, and states the scope of the defect. It does not write code.
Engineer	*proposes a patch*, into the substrate — a *different artefact* from the campaign's evidence. It cannot admit its own fix.
Supervisor	*gates resumption.* Evidence admission restarts only after the full ladder passes on the patched substrate, and the re-anchoring is recorded explicitly rather than absorbed.
Human	approves substrate changes. The campaign runs unattended; the thing it measures with does not change unattended.

The underlying principle is the one an experimentalist would recognise: *the instrument must not be adjustable by the experiment it is measuring.* Our situation is sharper than either published system's, because their measurement is an external wet-lab assay and ours is code written by the same project that is being judged.

9 What the campaign produces

Five artefacts, each a deliverable in its own right.

9.1 The ledger

Every claim sorted into **Established**, **Refuted**, **Structural limitation** or **Open**, with its evidence, sample count, confidence interval and the threshold set that produced the verdict. A mechanism reaches Established only if it answers what a biology paper would ask: remove it —

does the phenotype collapse? Is a minimal recipe containing it sufficient? Does it survive repeats and a range of settings?

9.2 Causal descriptions — "this spec gives this phenomenon"

For every surviving composition the Interpreter writes a structured account, attached to the record as a versioned analysis:

```
{ composition:  <structural hash + operator list + typed connections>,
  phenotype:    <class, with the metric vector that assigns it>,
  route:        "growth gated by the activator raises target volume locally;
                 oriented division builds a wall along the bud axis; T1 lets
                 the wall cells slide, so the tube extends instead of bulging",
  necessary:    [ops whose ablation collapses the phenotype],
  sufficient:   <the minimal sub-composition that still produces it>,
  falsified:    [alternatives tried and refuted, with run ids],
  scope:        <substrate, parameter ranges, seeds>,
  watcher:      <caption of the movie>,  agreement: <caption vs metrics> }
```

This is what makes the operator library reusable rather than merely catalogued, and it is what a reader of the eventual paper actually wants. It is generated per composition, continuously, not written up afterwards.

9.3 The operator backlog — atlas growth

Every escalation records an **operator request**: a mechanism the proposer wanted and the language could not express. Each is a candidate new operator and each is a talk bullet. Our expectation, from the one simulator decomposed in full, is that mechanics *saturate* against the existing algebra while topology and numerics contribute genuinely new contracts — and that saturation is itself the hypothesis under test. If genuinely new families keep appearing, the language is incomplete; if they stop, the compactness claim is supported.

9.4 Movies

Every promoted composition is rendered to the standard of the public gallery, captioned by the watcher, and archived beside its record. Captioning is on by default and is never disabled: the caption is simultaneously documentation, a semantic regression test, and the watcher's input. These are the primary evidence for a talk, and the loop produces them rather than a person producing them afterwards.

9.5 One differentiable figure

See §10.

10 Where the differentiable version enters

Mechanism search and parameter fitting must not run simultaneously, or neither conclusion is interpretable. The fit is *earned*, and the trigger is explicit:

> A composition that is **Established** but whose residual on a chosen objective persists *across the whole parameter basin* is handed to the differentiable fit. A residual that

survives every setting is evidence about a mechanism; a residual that disappears at some setting was never a mechanism question.

The objective is local in space and time. A loss spanning every cell over hundreds of frames unrolls, under reverse-mode differentiation, into a graph whose memory scales as cells $\times$ steps and rapidly becomes intractable. But the evidence constraining any single operator is usually confined to a small, transient region — the advancing front, the neck, the moment a wall forms. So the loss is placed exactly there: a region of interest over a few tens of steps. The mechanism is global and shared; the objective is not.

The natural objective here is the stress field. The tool that computes per-cell force, pressure and cortical tension is already torch-differentiable, and the campaign's discriminator Q is mechanical. "Minimise tube pressure while preserving aspect" is therefore both the biological question (is a low-stress tube reachable?) and a well-posed differentiable objective.

Deliverable. One figure: residual descent, with identifiable parameters distinguished from degenerate ones. A degeneracy found is a result — it says two mechanisms are indistinguishable under the current observable, which is a measurement question, and it goes back to the observable bank.

11 Pre-flight: eleven defects that must be closed first

The campaign will emit scientific claims, including claims that something is *impossible*. A false impossibility claim is worse than no claim, because it closes a question that should stay open. Defects 1–6 come from an adversarial audit of the simulation operators; 7–11 were found while writing this document, by reading the loop's own source with the same suspicion.

1. Two operators kept a private clock. They spaced themselves out while the engine *also* spaced them out; the schedules multiplied, so "every 2 steps" fired every 4. Division across every archived run happened at a fraction of the advertised rate. *Fix: the engine owns the clock; baselines re-measured and the re-anchoring stated, not absorbed.*

2. Adding a signalling step silently changed the time-step. The run script chose the time-step from which operators were present — recipes with a chemical signal ran fifty times finer. Since the campaign's most important edit is adding or removing signalling, every conclusion about whether signalling matters would have been confounded. *Fix: one time-step for the whole search; stability handled inside the operator that needs it.*

3. A silent mismatch produced a result we believed for days. Positions and connectivity were recorded on different strides; the analysis paired coordinates with another moment's connectivity and reported cells as inverted. Every physical cure failed — in hindsight, the clue. *Fix: equal length asserted, not clamped.*

4. An operator that does nothing still returns numbers. If a prerequisite is missing, the operator silently no-ops, the run finishes, and a score is produced. Harmless under hand-written recipes; under automatic search it is the most dangerous bug possible — the loop records "this mechanism cannot make tubes" when the mechanism never ran. *Fix: preconditions declared and checked; every operator reports whether it acted; a scheduled operator that never acted is an error, not a result.*

5. Some operators advertise capabilities they lack. One is labelled differentiable and discards gradients; another claims three dimensions while its validity check is two-dimensional. Labels that lie are worse than absent labels, because the search trusts them. *Fix: corrected and mechanically certified.*

6. Measurements can pass while the picture is wrong. A lumpy blob scored acceptably on every number we had, because the scores measured how far things stick out and a blob sticks out. *Mitigation: the watcher, plus the instrument gate below.*

7. The archive stores only the final frame. Each run computes several recorded frames and persists only the last, so every re-scoring is restricted to end-state observables. This contradicts the entire record: the decisive evidence was temporal — a protrusion that peaks mid-run then degrades, a pattern that floods only once growth starts, divisions arriving in synchronised waves. Worse, the artefact that cost us days is by construction a time-series defect, and an archive of final frames cannot be re-examined for it. *Fix before any long run: persist the full trajectory and a per-frame metric table. Storage is cheap; re-simulating weeks of cluster time to recover a time course we already computed and discarded is not.*

8. Impossibility is asserted by a hard-coded narrative. The strongest claim the system makes is currently produced by a low score plus an `if/elif` chain guessing a reason from absent operators. The reason is a template, not evidence; combined with defect 4 this is machinery for manufacturing confident false impossibility. *Fix: an impossibility claim requires (i) every scheduled operator verified to have acted, (ii) a minimum sample count with an explicit interval, (iii) a recorded, bounded falsification attempt — a deliberate search for any setting that does produce the phenotype — and (iv) a stated scope. Absent any of the four, the verdict is Open.*

9. Four samples cannot support a robustness claim. The default basin is two seeds by two parameter draws, so a rate is one of $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, and promotion thresholds are hard-coded constants — Established at ≥ 0.6 (three of four), structural below 0.2 (zero of four), necessary if ablation drops below 0.5. *Fix: a minimum sample count before promotion; a Wilson interval rather than a point rate; thresholds as named, recorded campaign parameters so a later re-analysis can vary them.*

10. The search accelerator is dead code in the long-run driver. Loop I accepts an optional surrogate that ranks candidate edits so the frontier expands best-first; the overnight driver never passes it. Every unattended round to date has been uniform breadth-first — the exact opposite of §5.

11. Per-round cost grows without bound. The driver deepens the search every four rounds and densifies the basin every twelve, with no ceiling.

The instrument gate

Before the first round, the metric bank must correctly separate a set of already-archived runs we have labelled by eye — a clean tube, a capped lobe, a flooded shell, a uniform sphere, a cauliflower. If the instrument cannot reproduce our own judgement on known cases, it is not ready to judge unknown ones. This is not a formality: three metrics in the existing bank have been caught lying, one of which moved a headline number by thirty per cent once corrected.

The validation ladder

One command, each level independently pass/fail, non-zero exit on any failure.

Level	Checks	Error class caught
Contract	every operator declares what it is and does	labels that lie
Unit	each operator reproduces its previous output	the refactor changed physics
Trajectory	a run reproduces its recorded history	schedule or ordering drift
Invariants	closed surface, no tangled or bow-tie faces	silent geometric corruption
Recording	positions and connectivity are equal length	the phantom-result bug
Metrics	scores match archived values	the analysis path drifted
Gradients	derivatives flow and are finite	the fit will fail later
Coverage	every operator exercised by some configuration	untested code presumed working
Description	the watcher's caption matches expectation	the numbers pass, the picture is wrong

One rung deserves comment. A face must not be self-intersecting, and we had checked validity by requiring positive area. *A bow-tie has positive area.* The check passed, the mesh was tangled, and a quantitative result about cell rearrangement was wrong — caught by watching a movie, not by any number. The lesson generalises well beyond meshes: a numerical invariant is not the geometric one you meant.

12 Operating it

The partition is cheap enough that wall-clock is not the binding constraint — attention is. The operating rules follow from that.

- **Read the ledger, not the runs.** The weekly unit of human attention is the ledger delta — what moved between the four classes — plus a montage of the week's best and the current operator backlog. Individual runs are opened only when the ledger points at one.
- **Verify against the world, never against a submission's reply.** A submission that reports failure may have succeeded, and one that appears to time out may already be queued; we have created duplicate jobs this way. The queue is the only ground truth. Generalised: an action's *reported* outcome is a hint; the world's state is the fact.
- **Every round is restartable.** Evidence is persisted as produced, exceptions caught per round, run identity a content hash — so a crash resumes without duplicating or losing evidence.
- **The configuration is written before the first round**, not discovered during it.

13 Status and the switch-on gate

The loop has run on gland branching and produced, unattended, a sequence that included one conclusion overturned by a better measurement — the strongest evidence that the evidence/interpretation separation is worth its cost. The Okuda operators exist, have produced a tube, and have produced the more interesting negative result that our tube is forced where the published one is grown. That is the question the campaign inherits, and §1 turns it into a measurable one.

The campaign switches on when four things are true:

1. the eleven defects are closed, each demonstrated by a test that fails loudly;
2. the archive stores trajectories and per-frame metrics, not final frames;
3. the metric bank has reproduced our own eye-labels on already-archived runs;
4. the supervisor's campaign configuration — objective, criteria, stopping rule, budget — is written down.

That gate is the point of this document. The system is now capable of running unattended for a long time and producing confident, well-formatted, entirely wrong conclusions. The defence is not more compute. It is a small number of checks that fail loudly, an archive that keeps raw evidence separate from its interpretation, a search that drops its losers instead of lovingly refining them, a supervisor that holds the goal nobody else can be trusted to hold — and the willingness to keep looking at the pictures.